Finite-Time Swarm Tracking: Numerical Results

All quantities are normalized. This report retains the arbitrary formation demonstration (Exp. 1), the finite-time convergence and coupling gain evaluation (Exp. 2), the star-formation safety and connectivity transient comparison across three representative cases (Exp. 3), and the Experiment 4 coupling–speed parameter sweep.

1 Model and parameters

The physical agents use

$$\dot{x}_i = u_i = \dot{q}_i - \delta_i - \mathrm{sig}^{\beta}(\delta_i) - \sum_j w_{ij} M(\delta_i - \delta_j), \qquad \delta_i = x_i - q_i,$$

where $\mathrm{sig}^{\beta}(z) = \mathrm{sign}(z)|z|^{\beta}$, $M = mI_2$, $\beta = 0.5$, and $w_{ij} = a_{ij}/(\|x_i - x_j\| + 0.2)$ with $a_{ij} = 1$ if $\|x_i - x_j\| < R_c$ $(i \neq j)$ and 0 otherwise. No centroid feedback, state reset, or controller smoothing is used. The ten centered star stations are

$$q_i(t) = r(t) + a(t)p_i^0, \qquad a(t) = a_\infty + (1 - a_\infty)e^{-0.9t}, \quad a_\infty = 1.65 - 0.25\kappa,$$
$$r(t) = [0.9t, 1.1\sin(0.35t)]^T, \qquad \dot{r}(t) = [0.9, 0.385\cos(0.35t)]^T,$$

with $p_i^0 = b_i[\cos\phi_i, \sin\phi_i]^T$, $\phi_i = \pi/2 + \pi i/5$, and $b_i = 2.8$ for even i and 1.22 for odd i, $i = 0, \ldots, 9$, centered so $\sum_i p_i^0 = 0$. The tightest station pair has distance $DQ_0 = 1.4342$, and the template diameter is $\ell_{H,0} = 5.3259$. Station velocities are analytic. The full horizon is $t_f = 12.0$, and the safety and connectivity transients are analyzed over $t \in [0, 2.5]$. The safety threshold is $d_{\mathrm{safe}} = 0.95$, and the communication radius is $R_c = 6.60$.

The initial tracking perturbation $\delta_i(0)$ is zero-mean $(\sum_i \delta_i(0) = 0)$, generated from Gaussian noise with $\sigma = 0.03$, with the closest template pair $(5, 7)$ nudged inward by 0.12 and the farthest pair $(0, 4)$ nudged outward by 0.22. This gives initial error norm $\|\delta_0\| = 0.3712$, effective distortion $\rho_{\mathrm{eff},0} = 0.4185$, and universal deadline $\overline{T}_{\mathrm{univ}} = 2\ln(1 + \sqrt{\|\delta_0\|}) = 0.9515$.

The time-varying geometric margin is

$$s(t) = \min\{a(t)DQ_0 - d_{\mathrm{safe}},\ R_c - a(t)\ell_{H,0}\}.$$

For coupling gain m, the effective rate is $\gamma = 1 + mw_{\mathrm{lower}}\lambda_2(L_H)$ with $w_{\mathrm{lower}} = 1/(R_c + \epsilon_w) = 1/6.8$ and $\lambda_2(L_H) = 10$. The distortion bound is

$$\rho_H(t) = \min\{\sqrt{2\overline{R}(t,\gamma)},\ \rho_{\mathrm{eff},0}e^{-\gamma t}\}, \qquad \overline{R}(t,\gamma) = \max\left\{0, \left(\frac{(1 + \gamma\sqrt{\|\delta_0\|})e^{-\frac{1}{2}\gamma t} - 1}{\gamma}\right)^2\right\}.$$

The required rate γ_{req} is evaluated over $\tau \in (0, \overline{T}_{\mathrm{univ}}]$ by maximizing $\frac{\ln(\rho_{\mathrm{eff},0}/s(\tau))}{\tau}$.

2 Formation demonstration

The original ring–wedge–star–arrow example uses smooth quintic blends and analytic station derivatives. It demonstrates tracking of reconfiguring formations; it makes no general collision-avoidance claim.

3 Finite-time tracking and coupling gain evaluation

Experiment 2 evaluates the safe-certified star formation maneuver ($\kappa = 3.0$, scale rate 0.90) across coupling gains $m \in \{0, 0.5, 2, 2.5, 5, 10\}$. Figure 2 shows the station tracking error $\|\delta(t)\|_2$, the formation disagreement $\|\eta(t)\|_2$,

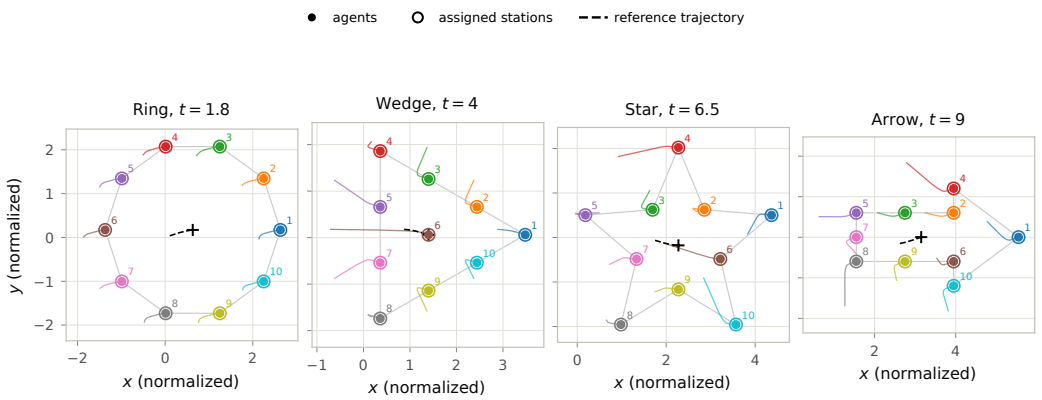


Figure 1: Moving reference and reconfiguring formations.

and the physical control command $\|u(t)\|_2$. In panels (a) and (b), the curves are cut at 10^{-4}, capturing the clean finite-time transient down to settling without plotting the post-convergence numerical discretization noise.

As coupling gain m increases, formation disagreement decays substantially faster, accelerating overall settling time to the 10^{-6} error threshold from 0.7683 s at $m = 0$ down to 0.1883 s at $m = 10$ (Table 1). All runs settle well within the theoretical universal deadline $\overline{T} = 0.9515$ s.

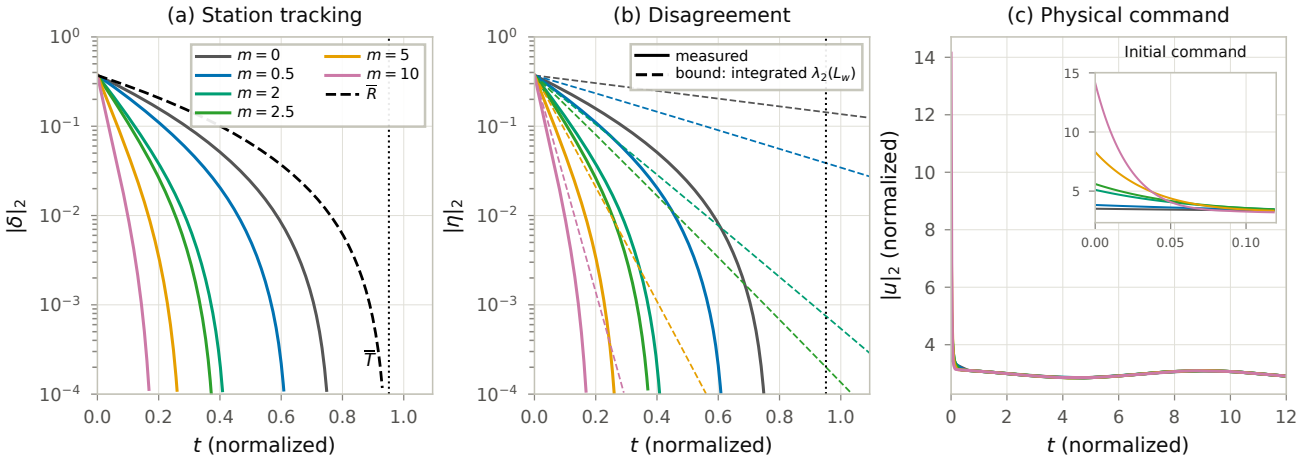


Figure 2: Experiment 2 finite-time tracking and disagreement across coupling gains m ($\kappa = 3.0$). (a) Station tracking $\|\delta\|_2$ against universal bound $\overline{R}$ and deadline $\overline{T}$, cut at 10^{-4}. (b) Disagreement $\|\eta\|_2$ against exponential rate bound, cut at 10^{-4}. (c) Physical control command $\|u\|_2$ with initial transient inset.

4 Star formation safety and connectivity transient

To evaluate certified safety and connectivity during formation transients, we examine three representative cases over $t \in [0, 2.5]$ s:

- **Case A (Certified Safe, $m = 2.5, \kappa = 3.0$):** With coupling gain $m = 2.5$, the convergence rate $\gamma = 4.6765$ exceeds $\gamma_{\text{req}} = 0.0430$, and the margin tube remains strictly positive throughout ($\min_t[s(t) - \rho_H(t)] = 0.0657 > 0$). In simulation, the swarm settles within $t_{\text{set}} = 0.3900$ s. Both safety ($d_{x,\min} = 1.1942 > d_{\text{safe}} = 0.95$, clearance 0.2442 m) and connectivity ($\ell_{x,\max} = 5.7444 < R_c = 6.60$, margin 0.8556 m) are preserved across the entire transient without any violation.

- **Case B (Min-Distance Breaks, $m = 0.0, \kappa = 4.5$):** With uncoupled agents ($m = 0.0$) undergoing rapid formation compression ($\kappa = 4.5$), the decay rate is only $\gamma = 1.0 < \gamma_{\text{req}} = 1.5888$. The uncertified tube dips below zero ($\min_t[s(t) - \rho_H(t)] = -0.1970$). In simulation, the swarm settles at $t_{\text{set}} = 0.7683$ s, but

Table 1: Experiment 2: Settling times and control metrics across coupling gains m under safe-certified star formation ($\kappa = 3.0$).

m	$t_{10^{-6}}$	$\overline{T}$	$\max \|u\|_2$	J_u
0	0.768333	0.951538	3.50984	0.32748
0.5	0.628333	0.831398	3.82752	0.438974
2	0.43	0.621194	5.10868	0.772248
2.5	0.39	0.576439	5.60358	0.883193
5	0.28	0.43254	8.30945	1.43756
10	0.188333	0.300258	14.1435	2.54577

a genuine collision violation occurs mid-transient: $d_x(t) < d_{\text{safe}}$ for $t \in [1.38, 2.50]$ s (duration 1.12 s), reaching a minimum distance of $d_{x,\min} = 0.7530$ m (a breach of 0.1970 m). Connectivity is preserved ($\ell_{x,\max} = 5.7444 < R_c$).

- **Case C (Connectivity Breaks, $m = 0.0, \kappa = 1.0$):** With uncoupled agents ($m = 0.0$) undergoing formation expansion ($\kappa = 1.0$), $\gamma = 1.0 < \gamma_{\text{req}} = 2.2654$. In simulation, the swarm suffers a genuine connectivity violation: $\ell_x(t) > R_c$ for $t \in [1.01, 2.50]$ s (duration 1.49 s), reaching $\ell_{x,\max} = 7.4562$ m (an excess of 0.8562 m). Pairwise separation remains strictly safe ($d_{x,\min} = 1.1942 > d_{\text{safe}}$).

Table 2: Experiment 3 outcomes: Star formation transient across Cases A, B, and C ($t \in [0, 2.5]$ s). Settling t_{set} is the time to reach error norm $\leq 10^{-6}$.

Case	m	κ	γ	γ_{req}	t_{set} [s]	$d_{x,\min}$	$\ell_{x,\max}$	Peak $\|u\|$	J_u	Outcome
A	2.5	3.0	4.6765	0.0430	0.3900	1.1942	5.7444	5.6036	0.8832	Certified; Safe, Connected
B	0.0	4.5	1.0000	1.5888	0.7683	0.7530	5.7444	4.8048	0.3275	Uncertified; Safety Violation
C	0.0	1.0	1.0000	2.2654	0.7683	1.1942	7.4562	3.8068	0.3276	Uncertified; Connectivity Violation

Figure 3 illustrates the 3-by-3 comparison of the transient responses. The left column shows the minimum pairwise distance $d_x(t)$ along with the analytical lower tube bound $d_q(t) - \rho_H(t)$ and the safety threshold $d_{\text{safe}} = 0.95$ m. The middle column plots the maximum pairwise distance $\ell_x(t)$ with the upper tube bound $\ell_q(t) + \rho_H(t)$ and the connectivity limit $R_c = 6.60$ m. The right column displays the geometric margin $s(t)$ and tube radius $\rho_H(t)$, highlighting certified ($s \geq \rho_H$) versus uncertified ($s < \rho_H$) regions.

5 Coupling–speed parameter map

Experiment 4 extends the same star template and initial physical state over the 4×4 grid $m \in \{0, 1, 2, 3\}$ and $\kappa \in \{0.18, 1.62, 3.06, 4.5\}$. Only coupling and compression speed vary; the reference, thresholds, backbone and 12-unit horizon are shared. For each point, the analytic condition $1 + m w_{\text{lower}} \lambda_2(L_H) > \gamma_{\text{req}}$ is compared with simulated separation and connectivity. The categories are 4 certified and observed safe/connected (A), 0 uncertified but observed safe/connected (B), and 12 with observed violations (C), comprising 4 cases with separation breaches at high compression ($\kappa = 4.5$) and 8 cases with connectivity loss at low compression/expansion ($\kappa \in \{0.18, 1.62\}$). There are no certified-case validation failures. The minimum observed separation is 0.752965; peak full command norms span 3.43483–7.31665.

The sufficient gain boundary is evaluated over $\kappa \in [0.18, 4.5]$ using the continuous-time margin calculation: for low $\kappa < 0.95$, rapid formation expansion renders the margin tube infeasible for certification; across intermediate speeds $\kappa \in [1.4, 4.0]$, the geometric margins are large enough that uncoupled tracking ($m = 0$) is certified safe; and for $\kappa > 4.1$, rapid compression requires positive coupling gain, reaching $m \geq 0.4004$ at $\kappa = 4.5$. The map classifies only its 16 simulated points; its colored cells do not certify unsampled parameter values. All saved adaptive solver nodes were audited, and the closest observed boundary case retained its classification under tenfold tighter tolerances and half maximum step.

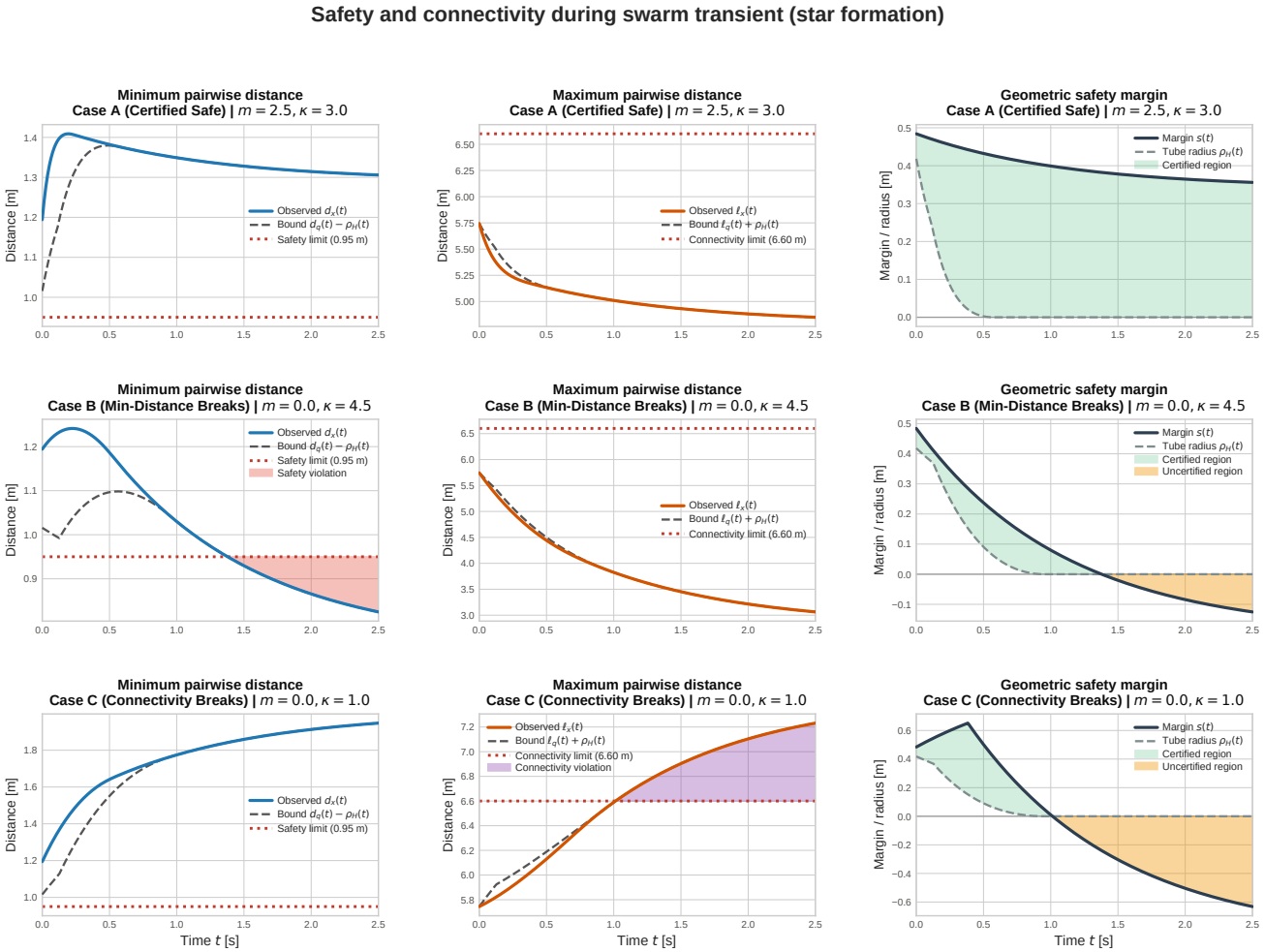


Figure 3: Safety and connectivity during swarm transient across the three star-formation cases ($t \in [0, 2.5]$ s). Columns: minimum pairwise distance $d_x(t)$ against $d_q(t) - \rho_H(t)$ and d_{safe} (left), maximum pairwise distance $\ell_x(t)$ against $\ell_q(t) + \rho_H(t)$ and R_c (middle), and geometric safety margin $s(t)$ against distortion radius $\rho_H(t)$ (right).

6 Reproduction and limits

Run `python3 run_experiments.py --experiment 2` to reproduce the coupling gain evaluation. Run `python3 run_experiments.py --experiment 3` from the repository root to reproduce the star-formation safety and connectivity transient experiment. Reproduce the parameter map with `python3 run_experiments.py --experiment 4 --grid-size 4 --workers 3`. The report source is `ACC_paper/numerical_results.tex`; compile it from `ACC_paper/` with `pdflatex numerical_results.tex`. The theoretical source is unchanged. The controller is first order; obstacle clearance, disturbances, actuator limits and Monte Carlo trials are not certified or studied here.

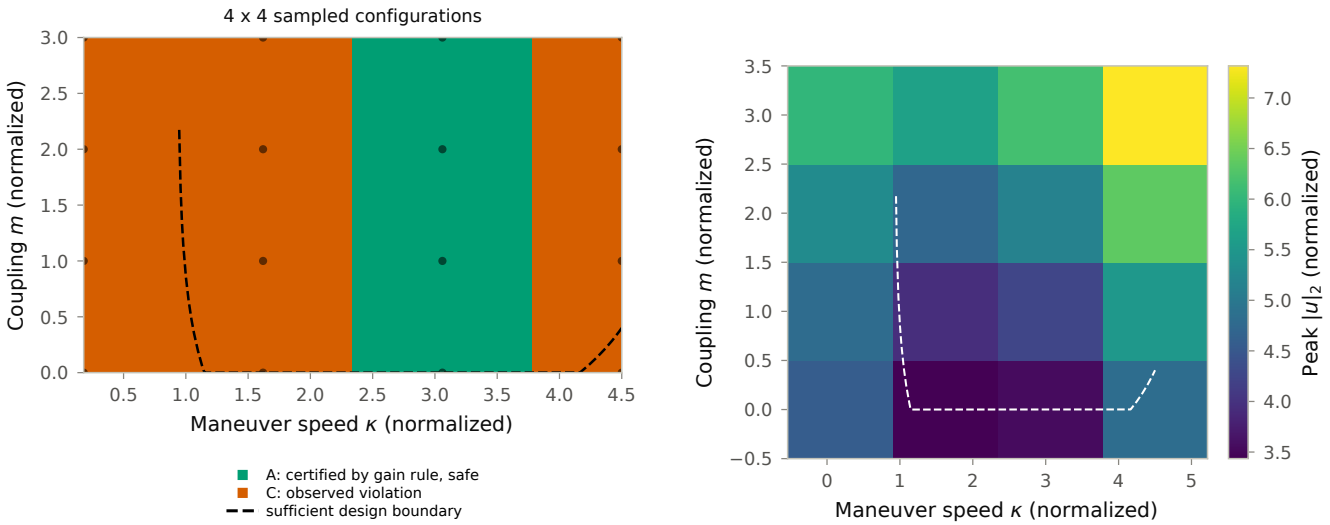


Figure 4: Experiment 4 pilot sweep. Left: sampled safety classifications and continuous-time sufficient gain boundary. Right: peak full physical command.